Daniel Zhang

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Education

Bachelor of Engineering, Mechatronics Engineering CO-OP — Hamilton, ON
McMaster University, Level III — Expected Apr 2028

- Notable Coursework: Analog & Digital Circuits, Statics, Dynamics, Electricity and Magnetism, Thermodynamics, Embedded Systems, Engineering Mathematics, Data Structures & Algorithms

Skills

Industry & Safety: Project Engineering, Process Flow Automation, Technical Writing, WHMIS, Lockout/Tagout

Design, Modeling & Hardware: CAD (SOLIDWORKS — CSWA Certified, Inventor, Fusion 360), FEA, NI Multisim, Embedded Systems, STM32, Soldering, 3D Printing, GD&T

Software: Python, C/C++, MATLAB/Simulink, Julia, SMath, Maple, FlexPDE, Microsoft 365, Power Automate

Experience & Involvement

Mechanical Sub-Team Member — Wheel Assembly — Hamilton, ON
McMaster Chem-E-Car — Sep 2026 – Present

- Designing and modeling a custom wheel assembly in SolidWorks for a $40 \times 30 \times 18$ cm vehicle, carrying up to a 500 mL water payload for the AIChE National Competition.
- Designing an independent double wishbone suspension to isolate dynamic weight transfer, stabilizing the vehicle chassis to prevent false triggers on optical stopping sensors.
- Collaborated with a team of 3 on an adjustable manual transmission, enabling on-site drivetrain tuning for target distances announced on competition day.

Nuclear Engineering Co-op Student — Cambridge, ON
Eclipse Automation — Jan 2026 – Present

- Authored FMEAs, FAT Plans, and System Design Descriptions for CSA N299.2/N299.3 rated tools, supporting milestones on the Pickering, Cernavodă, and Qinshan reactor refurbishment projects.
- Performed FEA and Allowable Strength Design (ASD) calculations for nuclear tooling on the Pickering (RFBR) project, validating structural integrity and safety margins.
- Delivered project engineering support for Design Verification Testing (DVT) of 4 specialized manufacturing tools, partnering with toolmakers, automation technicians, electricians, and clients to validate system performance against design requirements and facilitate customer sign-off.
- Facilitated CNSC regulatory approval for CNL's $100M SBWRS project by verifying 5,000+ technical requirements; built a Power Automate workflow to audit 1,000+ engineering standards and flag discrepancies, cutting review time from 5 days to 8 hours.

Projects

Ontario Nuclear Grid Analytics Dashboard | Live · GitHub — Oct 2026

- Built an end-to-end pipeline (Python → DuckDB → SQL → Streamlit) on 11.8M rows of IESO hourly generation, intertie, and price data for 193 generators (2019–2026).
- Wrote 16 SQL queries using window functions to detect 206 multi-day outages across Ontario's 18 reactors, label 1.16M reactor-hours by operating state, and measure how gas, hydro, and imports filled each gap.
- Quantified outage impacts on price and emissions; Darlington Unit 4's refurbishment added an estimated ∼64,000 t of CO_2 in its first two weeks as gas covered the lost output.

12-DOF Quadruped Robot | GitHub — Aug 2026

- Designed a 12-DOF quadruped in Autodesk Fusion with a print-ready chassis, integrating MG90S servos, a PCA9685 driver, and a Raspberry Pi Zero 2 W.
- Developed a custom Inverse Kinematics (IK) solver for stable omnidirectional gait patterns, tested in MuJoCo, with OpenCV camera integration for autonomous obstacle avoidance.

Rate-Adaptive Pacemaker Firmware | GitHub — Oct 2025 – Dec 2025

- Developed pacemaker firmware in Simulink/Stateflow for an NXP FRDM-K64F, generating embedded C via Embedded Coder to support AOO, VOO, AAI, and VVI pacing modes.
- Implemented rate-adaptive modes (AOOR, VOOR, AAIR, VVIR) using an FXOS8700 accelerometer, scaling pacing rate with patient activity via tunable threshold, reaction, and recovery times.